2.a) $V^{\pi}(s) = E_{\pi}[G_t / S_t = s]$

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$$

$$G_{t+1} = R_{t+2} + \gamma R_{t+3} + \gamma^2 R_{t+4} + \dots$$

$$\Rightarrow \gamma G_{t+1} = \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$$

$$\Rightarrow G_t = R_{t+1} + \gamma G_{t+1}$$

b) $v^{\pi}(s) = E\left[G_t \mid S_t = s, \pi\right]$

$= E\left[R_{t+1} + \gamma G_{t+1} \mid S_t = s, \pi\right]$

$= E\left[R_{t+1} + \gamma E\left[G_{t+1} \mid S_{t+1}, \pi\right] \mid S_t = s, \pi\right]$

$= E\left[R_{t+1} + \gamma v^{\pi}(S_{t+1}) \mid S_t = s, \pi\right]$

$= \sum_{a} \pi(a|s) \sum_{s', r} p(s', r \mid s, a)\left[r + \gamma v^{\pi}(s')\right]$